

Experiment No. 1

Doppler Effect using GNU Radio and ADALM-PLUTO SDR

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Aim

To design and implement a Doppler Effect detection system using GNU Radio and ADALM-PLUTO SDR that transmits a tone signal and receives Doppler shifted reflections from moving objects.

Objective

The objective of this experiment is to study the Doppler Effect using GNU Radio and ADALM-PLUTO SDR. The experiment aims to analyze frequency shifts caused by moving objects and observe real-time signal processing using SDR techniques.

Theory

When a radio wave reflects from a moving object, its frequency changes slightly. This phenomenon is called the **Doppler Effect**. If the target moves toward the transmitter, the received frequency increases, whereas motion away from the transmitter decreases the frequency.

$$f_d = \frac{2vf_c}{c}$$

Where:

Symbol	Meaning
f_d	Doppler frequency shift
v	Velocity of moving object
f_c	Carrier frequency
c	Speed of light

Table 1: Parameters used in Doppler Shift Equation

For this experiment, the carrier frequency is:

$$f_c = 3.5 \text{ GHz}$$

For an object moving at 3 m/s:

$$f_d = \frac{2 \times 3 \times 3.5 \times 10^9}{3 \times 10^8}$$

$$f_d \approx 70 \text{ Hz}$$

In this experiment, the transmitted signal reflects from a moving object such as a hand. Due to motion, the reflected signal experiences a slight frequency shift. Since the carrier frequency used is very high (3.5 GHz), even small movements generate measurable Doppler shifts. The received signal is filtered and processed to isolate these weak frequency variations.

Equipment Required

- ADALM-PLUTO SDR
- GNU Radio Companion (GRC)
- Antenna
- USB Cable
- Computer/Laptop

Software Requirements

- GNU Radio
- PlutoSDR Drivers
- libiio Support

Block Diagram

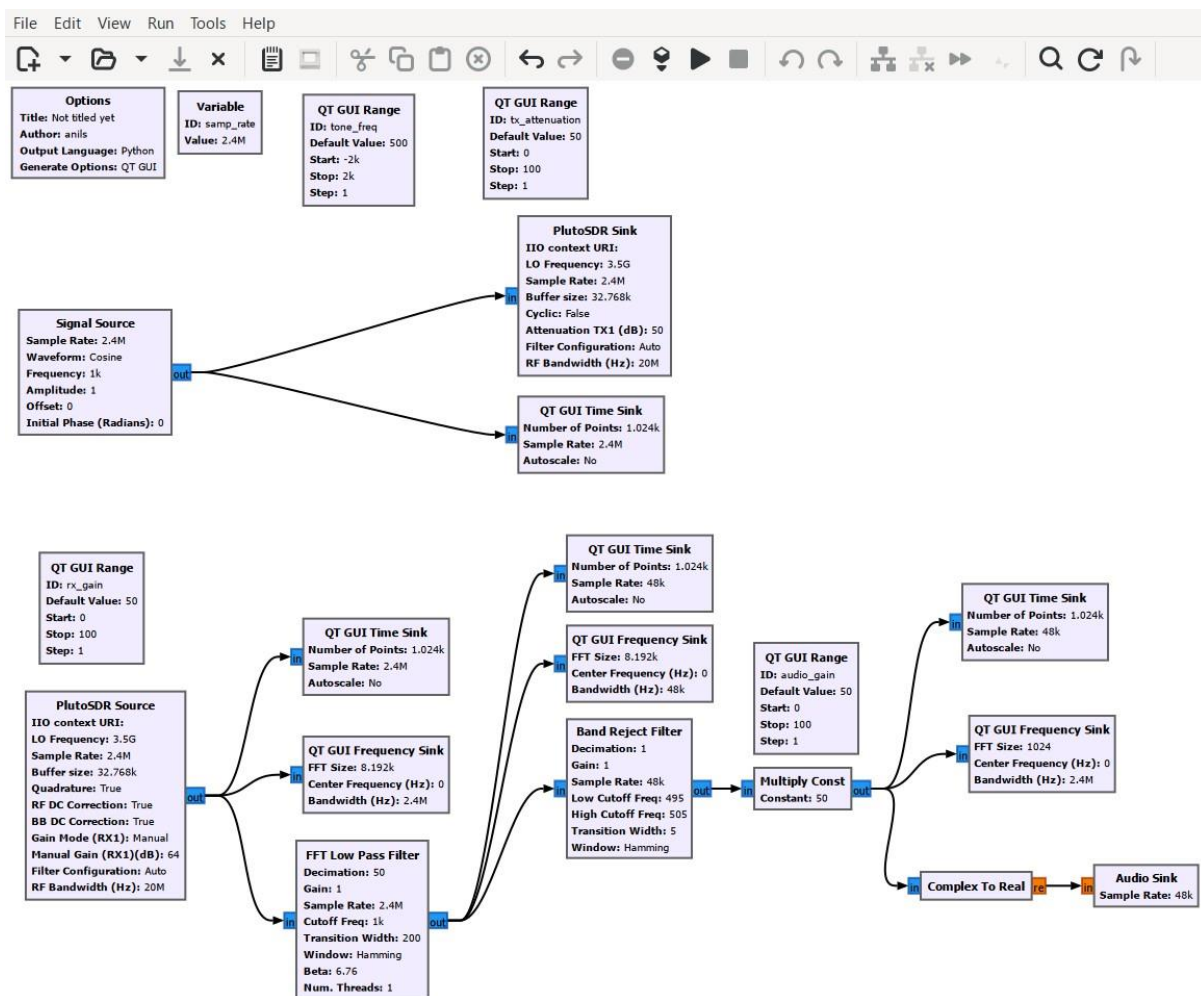


Figure 1: GNU Radio Doppler Effect Flowgraph

Flowgraph Description

The system contains two major sections: transmitter and receiver.

Transmitter Section

The transmitter section consists of a Signal Source block connected to the PlutoSDR Sink. A cosine signal of adjustable frequency is generated and transmitted at a carrier frequency of 3.5 GHz. The QT GUI Time Sink is used to monitor the transmitted waveform in real time.

Receiver Section

The receiver section consists of the PlutoSDR Source, Low Pass Filter, Band Reject Filter, Multiply Const block, Complex to Real block, and Audio Sink. The PlutoSDR Source receives both the direct transmitted signal and the reflected signal from moving objects. The Low Pass Filter reduces the bandwidth and decreases the sample rate to 48 kHz. After filtering, the Band Reject Filter removes the strong unshifted carrier component, allowing the weak Doppler shifted frequencies to become more visible. The Multiply Const block amplifies the weak reflected signal, while the Complex to Real block converts the complex signal into a real-valued signal suitable for audio output. Finally, the Audio Sink produces an audible representation of the Doppler shifted frequency.

Working Principle

The system transmits a continuous tone signal using the PlutoSDR transmitter. When the signal reflects from a moving object, the reflected wave undergoes a Doppler frequency shift. The receiver captures both the direct and reflected signals. A Low Pass Filter reduces bandwidth and sample rate, while a Band Reject Filter removes the strong direct transmission component. The remaining Doppler shifted signal is amplified and converted into an audible signal using the Audio Sink.

Parameters

Parameter	Value or Range
Sample Rate	2.4 MS/s
Tone Frequency Range	-1 kHz to 1 kHz, default 500
RX Gain	0 to 70, default 64
RX Gain Mode	Manual
Pluto LO Frequency	3.5 GHz
TX Attenuation	0 to 100, default TBD
Audio Gain	0 to 100, default 50

Table 2: System Parameters

Filter Properties

Parameter	Value or Range
LPF Decimation	50
LPF Cutoff Frequency	$2 \times \text{tone}$
LPF Transition Width	250 Hz
LPF Window	Hamming
BRF Low Cutoff Frequency	tone - 5 Hz
BRF High Cutoff Frequency	tone + 5 Hz
BRF Transition Width	5 Hz
BRF Window	Hamming

Table 3: Filter Configuration

Procedure

1. Open GNU Radio Companion.
2. Create variables:
 - samp.rate = 2.4M
 - tone_freq
 - tx_gain
 - rx_gain
 - audio_gain
3. Add QT GUI Range blocks for tone frequency, TX gain, RX gain, and audio gain.
4. Add a Signal Source block and configure it for cosine waveform generation.
5. Connect the Signal Source to the PlutoSDR Sink.
6. Configure the PlutoSDR Sink with LO Frequency = 3.5 GHz and Sample Rate = 2.4 MS/s.
7. Add QT GUI Time Sink to monitor the transmitted waveform.
8. Add PlutoSDR Source block for receiving signals.
9. Configure the receiver with manual RX gain mode.
10. Add Time Sink and Frequency Sink for received signals.
11. Add a Low Pass Filter with decimation factor 50.
12. Add a Band Reject Filter to suppress the direct carrier component.
13. Add Multiply Const block for amplification.
14. Convert the complex signal into real signal using Complex to Real block.
15. Add Audio Sink with sample rate 48 kHz.
16. Run the flowgraph and move a hand or object near the antenna.
17. Observe Doppler shifts in the frequency spectrum and hear audio variations.

Observation Table

Object Motion	Frequency Shift	Observation
Towards Antenna	Positive Shift	Increase in pitch
Away from Antenna	Negative Shift	Decrease in pitch
No Motion	No Shift	Constant tone

Table 4: Observed Doppler Shifts

Result

The Doppler Effect was successfully demonstrated using GNU Radio and ADALM-PLUTO SDR. Frequency shifts caused by moving objects were detected, processed, and converted into audible signals.

Applications

The Doppler Effect is widely used in radar systems, speed detection devices, motion sensing applications, medical ultrasound imaging, traffic monitoring systems, and gesture detection technologies.

Advantages

This system provides real-time signal processing with a flexible SDR platform. It is low-cost, easy to modify, and suitable for practical radar experimentation using GNU Radio.

Limitations

Small Doppler shifts can be difficult to detect in noisy environments. Signal leakage between transmitter and receiver antennas may also affect system performance.

Future Scope

This system can be extended for gesture recognition, contactless monitoring, intelligent radar sensing, and IoT-based motion detection applications.

Conclusion

This experiment demonstrates how SDR technology can be used to study Doppler frequency shifts. GNU Radio provides a flexible platform for designing real-time radar and motion detection systems using PlutoSDR hardware.

References

1. GNU Radio Documentation
2. Analog Devices PlutoSDR Documentation
3. SDR Fundamentals
4. Wireless Communication Principles